

CO2 AT2 - Class Test

1. a) Data Mining, KDD & Preprocessing

Data Mining: The process of discovering useful patterns, correlations, & knowledge from large datasets using statistical, mathematical & ML tech.

Two basic data mining tasks:

(1) classification: Assign data to predefined categories.

Ex: Spam vs not spam

(2) clustering: Group similar data points without predefined labels.

Ex: Customer segmentation.

Steps involved in KDD

(1) Data Selection

(2) Data preprocessing

(3) Data transformation

(4) Data Mining

(5) Interpretation

Role of data processing:

Data is usually noisy, incomplete, & inconsistent. Preprocessing improves data quality. Without it, mining gives inaccurate patterns & models. It directly impacts accuracy & efficiency of mining algorithms.

b) Data cleaning

The process of detecting and correcting inaccurate, incomplete, or irrelevant data to improve quality.

Two common data quality issues:

(1) Missing values

(2) Noisy data

a) Data integration: It is the process of combining data from multiple sources into a single, consistent dataset for analysis.

Challenges:

* Data redundancy and data inconsistency.

Handling challenges:

* Remove duplicate records to reduce redundancy

* Standardize formats & resolve conflicting values to eliminate inconsistency

b) Covariance Analysis: It measures how two variables change together. It indicates whether the variables increase or decrease together.

Types:

(1) positive covariance:

(2) Negative covariance

- Suppose two stocks A & B have the values:

$A = (2, 3, 4, 5, 6)$ $B = (5, 8, 10, 11, 14)$

The covariance of two stocks is +ve (+4). A positive covariance indicates that both stocks tend to rise & fall together. Therefore, they are positively related & not independent.

(i) Database: Stores large amounts of data

(ii) Database server: Retrieves & manages data for mining

3) Data Mining Engine: Performs tasks like classification, clustering & analysis

4) Pattern evolution module: Identifies useful & interesting patterns

5) Knowledge base: Stores domain knowledge & constraints

6) Graphical user interface (GUI): Allows users to interact with the system & view results.

Covariance for:

$$x = \{4, 5, 6, 7, 9\} \quad y = \{6, 3, 7, 5, 6\}$$

Mean of $x = 6.2$

Mean of $y = 5.8$

Covariance $= -0.36$

* The covariance is -0.36 , indicating a slight negative relationship b/w x & y

b) Major steps in data preprocessing

(1) Data cleaning: Removes missing values and noise

(2) Data integration: Combines data from multiple sources

(3) Data transformation: Convert data into a suitable format

(4) Data reduction: Reduces data size while preserving information

(5) Data discretization: Convert continuous values into intervals.

5a) Perform equal-frequency binning.

Given data: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Divide into 3 bins:

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Equal-frequency binning places an equal no. of data values in each bin, helping reduce noise.

b. Data discretization: It is the process of converting continuous data into a finite no. of intervals.

Importance:

- + Reduce data size
- + Removes noise
- + Improves mining efficiency
- + Simplifies analysis

Comparison:

Equal width Binning

Equal-frequency Binning

Equal Interval size

Equal no. of record

Simple to implement

Better for skewed data